

Radiotherapy for oesophageal cancer

Radiotherapy uses high-energy rays to destroy oesophageal cancer cells.

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What is radiotherapy?

Radiotherapy uses high-energy rays called radiation to treat cancer. It destroys cancer cells in the area where the radiotherapy is given.

Some normal cells in the area being treated can also be damaged by radiotherapy. This can cause [side effects](#). As the normal cells recover, the side effects usually get better.

Radiotherapy for [oesophageal cancer](#) can be given as external beam radiotherapy or internal radiotherapy (brachytherapy).

External beam radiotherapy for oesophageal cancer

[External beam radiotherapy](#) is usually given in combination with chemotherapy to treat oesophageal cancer. This is called [chemoradiation](#).

For early stage oesophageal cancer, you might have chemoradiation:

- before surgery, to shrink the tumour
- instead of [surgery](#), with the aim of curing the cancer.
- instead of surgery if you are not well enough or choose not to have surgery
- as part of a [clinical trial](#).

If the cancer is more advanced, you may have radiotherapy on its own to shrink the tumour and help control symptoms.

Chemoradiation for oesophageal cancer

Chemoradiation is used to treat squamous cell and adenocarcinoma of the oesophagus. The chemotherapy drugs can make cancer cells more sensitive to radiotherapy. This can help the radiotherapy work better.

These chemotherapy drugs include:

- [fluorouracil \(5FU\)](#)
- [capecitabine](#)
- [cisplatin](#)
- [carboplatin](#)
- [paclitaxel](#).

Having chemotherapy and radiotherapy together can make the side effects of treatment worse. It may not be suitable for people who have other health problems. Your cancer doctor or specialist nurse can give you more information about [chemoradiation and its possible side effects](#).

Planning your radiotherapy

You will have a hospital appointment to [plan your treatment](#).

Feeding tube

Sometimes, radiotherapy can lead to [difficulty swallowing](#), which can make it hard to eat. You may have a feeding tube put into the stomach before the radiotherapy starts. This helps make sure you are getting enough to eat. There are different types of feeding tube:

- nasogastric feeding tubes are put into the nose and passed down the oesophagus into the stomach
- gastrostomy feeding tubes are passed through the skin into the stomach.

Liquid food can be given through the tube. You can go home with it in place. You, or any carers, can be shown how to manage the feeding tube at home with support from a dietitian or community nurse.

Your cancer doctor can talk to you about whether you might need a feeding tube.

We have more information about [feeding tubes and nutritional support](#).

Having radiotherapy treatment

Radiotherapy is normally given as a number of short, daily treatments in a hospital radiotherapy department. A radiotherapy machine aims high-energy rays at the area of the body being treated. You might hear this type of radiotherapy being called [external beam radiotherapy](#).

You usually have radiotherapy as an outpatient. You sometimes have it during a hospital stay. Usually, each appointment takes about 10 to 30 minutes. Your radiotherapy team will explain how many treatments you will have and when you will have them. Your course of treatment may last up to a few weeks.

The person who operates the machine is called a radiographer. They give you information and support during your treatment.

We have more information about having [external beam radiotherapy](#).

Support from Macmillan

Macmillan is here to support you. If you would like to talk, you can:

- Call the Macmillan Support Line for free on [0808 808 00 00](tel:0808 808 00 00) to talk to a cancer information specialist.
- [Chat to our specialists online](#).
- Visit our [oesophageal cancer forum](#) to talk with people who have been affected by oesophageal cancer, share your experience, and ask your questions.

Side effects of radiotherapy for oesophageal cancer

Radiotherapy can cause side effects in the area of your body that is being treated. You may also have some general side effects, such as [feeling tired](#).

Sometimes side effects get worse for a time during and after you have finished radiotherapy, before they get better.

Your cancer doctor, specialist nurse or radiographer will tell you what to expect. They will give you advice on what you can do to manage side effects. If you have any new side effects or if side effects get worse, tell them straight away.

Skin changes

The [skin](#) in the area that is treated may:

- become dark or red, depending on your skin type
- feel sore or itchy.

Your radiographer or specialist nurse will give you advice on taking care of your skin. If your skin becomes sore or itchy, or changes colour, tell them straight away. They can give you advice and treatments if needed.

Skin reactions should get better within 4 to 6 weeks of treatment finishing.

During your treatment, you are usually advised to:

- wear loose-fitting clothes made from natural fibres, such as cotton
- wash your skin gently with soap and water and gently pat it dry
- avoid rubbing the skin
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avoid very hot things, for example heating pads

- avoid cooling pads but these may be helpful in some situations so speak to your team about using these first
- avoid wet shaving
- avoid hair-removing creams or products, including wax and laser treatment
- follow your radiotherapy team's advice about using moisturisers and deodorants
- protect the treated area from the sun.

Sore throat and difficulty swallowing

Towards the end of treatment, radiotherapy can make your throat or oesophagus painful. You may not be able to swallow properly for a while. It can help to [eat soft foods until the pain improves](#). [Difficulty swallowing](#) and pain can last for a few weeks after radiotherapy finishes. Your cancer doctor or radiographer can give you medicines to help.

If swallowing is painful, it may mean you are not able to eat enough. If this happens, you may need to speak with a dietitian. They can talk to you about how you can supplement your diet with high-calorie, nutritious drinks.

Dry mouth

Sometimes, radiotherapy to the upper oesophagus reduces the amount of spit (saliva) the salivary glands make. This does not usually last long, but for some people it may be permanent. It may [make your mouth dry](#), which can make eating difficult.

Loss of appetite

During radiotherapy, you may not feel like eating very much. But it is important to try to maintain your weight. Your dietitian can give you advice on ways to build up your diet and [maintain or gain weight](#).

Feeling sick

Radiotherapy and chemotherapy can cause you to [feel sick \(nausea\) and be sick \(vomit\)](#). This can usually be treated with anti-sickness drugs (anti-emetics). Your cancer doctor can give you these.

Hair loss

When radiotherapy is used to treat oesophageal cancer, it can affect hair in the area being treated. If you have hair on your chest, [some of it may fall out](#).

Tiredness

Radiotherapy can make you feel [tired](#). In some people, this continues for several months after treatment finishes. During treatment, you will need to rest more than usual, especially if you have to travel a long way for your treatment each day. But gentle exercise, such as short walks, can help with energy levels. After treatment finishes, try to slowly increase your activity.

Late effects of radiotherapy to the oesophagus

Sometimes radiotherapy for oesophageal cancer can cause long-term side effects. These are side effects that:

- start during treatment but do not go away
- develop months or years after treatment finishes.

Most people do not get late effects. But if you notice any symptoms, always tell your cancer doctor or specialist nurse.

Difficulty swallowing

Radiotherapy can cause some damage to the tissue in the oesophagus. This can make it difficult to swallow. You may be offered treatment to stretch the oesophagus to make swallowing easier.

We have more information about [treating swallowing problems for oesophageal cancer](#).

Effects on the lungs

The lungs may become inflamed, or the tissue can become scarred (fibrosis). If you have a cough or are [short of breath](#), tell your cancer doctor or specialist nurse.

Effects on the heart

Radiotherapy can cause [heart problems](#) if the heart area is being treated. If you were told the radiotherapy may affect your heart, you can reduce the risk by making healthy lifestyle choices.

Effects on the spleen

Chemoradiation may damage the spleen if it is used to treat tumours that are close to the stomach. The spleen is an organ in the upper part of the abdomen, on the left-hand side. It helps us to fight infection and removes old and damaged blood cells from the bloodstream. Your doctor may talk to you about the need for immunisations and long-term antibiotics after your treatment.

We have more information about [late effects of cancer treatment](#).

Internal radiotherapy (brachytherapy) for oesophageal cancer

[Brachytherapy](#) uses radioactive implants such as seeds, pellets, wires or plates that are put near or inside the tumour. The radioactivity only affects tissue that is very close to the implant. This means the tumour is treated, but healthy areas around it get much less radiotherapy. Areas of the body that are further away are not affected at all.

The implants are left in place for 10 to 30 minutes, depending on the radiation dose being given.

Brachytherapy is not suitable for everyone. It is usually only given if the cancer has come back in the oesophagus after external radiotherapy or chemoradiation. Sometimes brachytherapy is given to help control the cancer and improve symptoms such as [difficulties with swallowing](#).

Internal radiotherapy is given through a nasogastric (NG) tube. A thin, flexible, plastic tube is placed up the nose, down the back of the throat, and into the stomach. A machine then sends radioactive pellets into the tube.

Your cancer doctor can explain more about how this treatment is given.

Internal radiotherapy can cause some pain when you swallow. This may start a few days after treatment starts, but usually goes away a few weeks after treatment finishes. Your cancer doctor can recommend liquid medicines to help.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our oesophageal cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

European Society for Medical Oncology. Oesophageal Cancer: ESMO clinical practice guidelines for diagnosis, treatment and follow up. 2022. Available from www.esmo.org/guidelines/guidelines-by-topic/gastrointestinal-cancers/oesophageal-cancer [accessed July 2023].

National Institute for Care and Health Excellence. Oesophago-gastric cancer: assessment and management in adults NICE guideline [NG83]. 2018. Available from www.nice.org.uk/guidance/ng83 [accessed July 2023].

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Date reviewed

Reviewed: 01 November 2024 | Next review: 01 November 2027

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to



Oesophageal cancer forum

An anonymous network of people affected by oesophageal cancer which is free to join. Share experiences, ask questions and talk to people who understand.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help